HABIBUL RAHMAN QALBI

AI Engineer | Backend Engineer (Python) | LLM Agents, RAG, FastAPI

Surabaya, Indonesia (UTC+7) | habi6799@gmail.com | linkedin.com/in/upchunk | github.com/upchunk | upchunk.github.io

Open to: remote roles worldwide | relocation to NZ, AU, SG, MY, TW, JP, US | contract and freelance projects

PROFESSIONAL SUMMARY

AI Engineer and Python Backend Engineer with 4+ years building production software, about 3 of them shipping LLM agents and RAG services with FastAPI. I make AI systems cheaper, faster and more accurate, and measure the result: **up to 70% lower cost** and **up to 4x faster** than a baseline tool-calling agent, and **~60% higher reasoning accuracy** after re-engineering retrieval. Working remotely since 2023 for a Taipei-based company and its enterprise customers, including Advantech, directly with C-level leaders.

TECHNICAL SKILLS

LLMs & GenAI: OpenAI, Anthropic (Claude), Google GenAI, VoyageAI, Pydantic AI, prompt and context engineering, function / tool calling, structured outputs, semantic and prompt caching, LLM evaluation, AI safety (PII scrubbing, prompt-injection prevention, moderation)

Agentic AI & RAG: AI agents, multi-agent orchestration, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, embeddings, semantic search, reranking

Backend & APIs: Python, FastAPI, Django, Django REST Framework, REST APIs, microservices, async processing, Celery, RabbitMQ, WebSockets, MQTT, JWT authentication

Data & Vector Stores: PostgreSQL, MongoDB, Redis, Pinecone, ChromaDB, SQL, Apache Airflow, web scraping (Playwright), PDF parsing

DevOps & Cloud: Docker, CI/CD, Git, Azure, MinIO

Frontend: React, Next.js, TypeScript, Redux Toolkit, RTK Query, Tailwind CSS, Material UI, Three.js

ML / Deep Learning: TensorFlow, CNNs, U-Net, computer vision, image classification and segmentation

Languages: English, Indonesian

PROFESSIONAL EXPERIENCE

AI Engineer

Nov 2023 - Present

Airconcur (Growise) | Taipei, Taiwan | Remote

Joined as a freelance AI Module Developer; hired full-time in May 2024.

- Architect and ship agentic and RAG-based AI microservices in Python (FastAPI) with multiple LLMs and embedding models, working directly with C-level leaders.
- Cut operating cost by **up to 70%** and made services **up to 4x faster** than a baseline tool-calling agent through semantic, prompt and tool-output caching, data canonicalization, and context and memory management.
- Re-engineered the retrieval layer (Voyage AI embeddings, reranking, data cleanup and de-duplication): **~30% lower latency** and **~60% higher reasoning accuracy** vs. earlier versions.
- Designed the agent architecture for TAM AI, an AI go-to-market platform whose agents research companies, decision-makers and buying signals across multiple data sources: multi-step reasoning, tool calling and structured outputs.
- Integrated Playwright browser automation, web scraping and PDF parsing into agent workflows so agents reason over clean, structured data.
- Built containerized, multi-LLM RAG modules with isolated per-user indexes to prevent cross-user data leakage, plus AI safety layers: PII scrubbing, prompt-injection prevention and moderation middleware.
- Delivered AI services for enterprise customers including Advantech (document generation, conversational assistants, transcription, image search); **2x faster builds** after reworking CI/CD.

Full Stack Developer / Software Engineer

May 2023 - May 2024

PT Mandiri International Technology | South Jakarta, Indonesia

- Owned the full project lifecycle, from requirement gathering with the project director to deployment.
- Built customer-facing web apps (React, Next.js) and a cross-platform mobile app on a Django REST backend with Celery workers.
- Engineered IoT device-backend communication over HTTP and MQTT: **50% higher pipeline performance** vs. a standard MQTT pipeline (parallel input-stream management) and **70% less device bandwidth** (multi-device connection routing).
- Added authenticated data reporting for site monitoring; tuned ORM and queries to handle **100+ concurrent requests**.

Full Stack & Backend Developer (Healthcare AI)

Aug 2022 - Mar 2023

Institut Teknologi Sepuluh Nopember (ITS) | Freelance | Surabaya, Indonesia

- Built the React-Django app and backend serving the research team's U-Net (TensorFlow) model for ultrasound deep-vein segmentation, with 3D visualization in Three.js (research prototype).
- Developed a model handler and preloader to reduce real-time inference latency.
- Replaced legacy auth with JWT and refactored API viewsets on the iBrain2U platform.

Full Stack Web Developer

Apr 2022 - Apr 2023

LnData Inc. | Surabaya, Indonesia

- Built React / Django products, secure APIs and Apache Airflow data pipelines for social-media data platforms, working with multinational teams.
- Doubled the capabilities of a data-visualization tool while improving its UX and performance.
- Managed deployment servers, monitored system load and resolved security issues.

EDUCATION

B.Eng. in Computer Engineering

2017 - 2021

Institut Teknologi Sepuluh Nopember (ITS) | Surabaya, Indonesia | GPA 3.25 / 4.00

- **Focus:** Artificial Intelligence, Microprocessor and Microcontroller Systems
- **Thesis:** CNN-Based Handwashing Motion Classification (published, see below)

PUBLICATION

Habibul Rahman Qalbi, Eko Mulyanto Yuniarno, Reza Fuad Rachmadi. "Klasifikasi Gerakan Cuci Tangan Berbasis Convolutional Neural Network (CNN)" [CNN-Based Handwashing Motion Classification]. *Jurnal Teknik ITS*, 10(2), 2021. doi.org/10.12962/j23373539.v10i2.75377

- Trained an EfficientNet-B0 classifier (TensorFlow) on a cleaned and extended video dataset; recognized all 12 handwashing steps in testing, with moving-average smoothing to stabilize predictions. Demo and case study: upchunk.github.io